

Senior Research Project

Initial Proposal

Due: Monday, April 20, 2026, by 2:10 PM

The initial proposal is the first checkpoint in your Senior Research Project. This document allows you to demonstrate that your research topic is thoughtful, feasible, and grounded in credible research. Your proposal will help faculty and staff determine whether your project is ready to move forward.

Your proposal should be approximately 2 pages in length (not including references) and include the following sections.

Submission Instructions: Your proposal must be **printed and physically dropped off in Dr. Valenti's classroom, room 319, by 2:10 PM on April 20. Digital submissions will not be accepted.** Be sure to also include your [signed acknowledgment form](#) when you submit your proposal.

1. Student Information

Include the following at the top of your proposal:

- Student name(s)
- Proposed project title
- Research pathway (choose one):
 - Empirical Research
 - Invention / Engineering Design
 - Research Assistantship
 - AP Seminar & AP Research
- Intended college major or field of interest (optional)

2. Introduction

This section should articulate the topic and importance of your research.

Include:

- Background information on your topic
- A clear explanation of the problem or question you want to investigate
- Why this topic is interesting, important, or relevant
- Any real-world applications of the topic
- Summary of what previous research has discovered
- Identification of a gap in knowledge or a problem that still needs investigation

Strong proposals typically cite 3-5 credible sources

3. Research Question

Clearly state the central question your research will attempt to answer.

Example formats:

- How does _____ affect _____?
- Can _____ improve _____?
- What factors influence _____?

4. Hypothesis or Project Goal

Depending on your identified pathway:

A. Empirical Research. State a testable hypothesis.

Example: If _____ is changed, then _____ will occur because _____.

B. Engineering / Invention. State the design goal or problem you will attempt to solve.

Example: The goal of this project is to design a prototype that _____.

C. Research Assistantship. Explain the research role or project you will contribute to

D. AP Seminar & AP Research. A two-year program focused on inquiry, academic writing, and formal presentation around a topic of personal interest.

5. Proposed Methods

Explain how you plan to conduct your research.

Your description should include:

- Type of study (experiment, design project, analysis, etc.)
- Variables (if applicable)
- Materials or tools you may need
- Data you plan to collect
- Approximate sample size or number of trials
- How you will analyze results (graphs, statistical tests, etc.)

Your methods do **not** need to be perfect yet, but they should show that you have thought about how the research will be conducted.

6. Expected Outcomes / Impact

Briefly explain:

- What you expect to find, create, or learn
- Why your findings might matter
- Possible real-world applications

7. References (APA Format)

Include at least 3 credible sources, such as:

- Peer-reviewed journal articles
- Academic books
- Scientific databases (Google Scholar, PubMed, JSTOR, etc.)

Citations should be in APA format.

8. Prior STEM Research Abstract

To receive project approval, all students must submit:

- A 250-300 word abstract summarizing their 9th/10th-grade STEM research project

This abstract must include:

- Research question or design goal
- Methods used
- Key results or outcomes
- What was learned